Modular versus Monolithic Neural Architectures for Gene Regulatory Network Inference: An Empirical Comparison Under Parameter-Matched Conditions

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Abstract—Gene regulatory network (GRN) inference — determining which transcription factors regulate which target genes from single-cell RNA-sequencing data — is a core problem in computational biology. A fundamental architectural choice is whether to encode transcription factor (TF) and gene representations independently (two-tower/dual-encoder) or jointly (cross-encoder). We present a parameter-matched comparison of both architectures on a human brain single-nucleus RNA-seq dataset (39K TF–gene pairs, 5.58M parameters each) under balanced (1:1) and moderately imbalanced (5:1) negative-to-positive sampling ratios. The cross-encoder achieves an AUROC of 0.9040 versus 0.8097 for the two-tower under balanced training — a gap of 9.4 points consistent across five random seeds. A no-product cross-encoder ablation (input without element-wise embedding product) achieves AUROC 0.8923, attributing ≈88% of the 9.4-point gap to joint encoding and ≈12% to the multiplicative interaction term. Under 5:1 negative sampling, the two-tower AUROC degrades by 6.6 points (0.8097 → 0.7434) while the cross-encoder AUROC is stable (0.9040 → 0.9150); both architectures exhibit AUROC and F1 degradation under imbalance, consistent with cosine similarity scoring being more sensitive to class imbalance than MLP-based scoring on this dataset. A structured neuron pruning analysis across three random seeds shows that post-hoc AUROC retention never falls below 99.96% of baseline at any tested sparsity level (minimum at 10% sparsity; retention equals or exceeds baseline at all higher sparsity levels), with 51 of 512 neurons per tower sufficient to match baseline AUROC without fine-tuning. All findings are on a single human brain dataset and are qualified to that experimental context.

Index Terms—gene regulatory network inference, two-tower architecture, cross-encoder, dual-encoder, neural network pruning, single-cell RNA-seq, representation learning, link prediction, class imbalance

I. Introduction

Gene regulatory network (GRN) inference — the problem of identifying which transcription factors (TFs) regulate which target genes — is a fundamental challenge in computational biology. Understanding transcriptional regulatory relationships is essential for deciphering developmental programs, disease mechanisms, and therapeutic targets. With the availability of large-scale single-nucleus RNA-sequencing (snRNA-seq) data, machine learning methods have emerged that cast GRN inference as binary link prediction: given a candidate TF–gene pair and their expression profiles, predict whether a regulatory relationship exists [1], [2].

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A central architectural question in link prediction is whether entity representations should be computed independently (a two-tower or dual-encoder model [3], [4]) or jointly (a cross-encoder [5], [6]). Two-tower models score interactions via cosine similarity of independently computed encodings, enabling efficient inference at scale. Cross-encoders process pairs jointly, allowing arbitrary interaction features to be learned before scoring. In information retrieval, cross-encoders consistently outperform two-tower models when precise ranking quality is needed [5], [6], but whether this advantage holds in computational biology — where entities have distinct biological identities and the regulatory signal is low-dimensional — has not been established under controlled, parameter-matched conditions.

Three open questions motivate this work: (1) Does a cross-encoder achieve higher discriminative AUROC than a parameter-matched two-tower for TF–gene link prediction? (2) Is the two-tower's cosine similarity scoring more sensitive to class imbalance than the cross-encoder's MLP scoring? (3) Does the trained two-tower actually utilize its nominal hidden-layer capacity, or is much of it redundant?

This paper makes three contributions:

- C1: A parameter-matched comparison (5.58M parameters each) of two-tower and cross-encoder MLPs on human brain snRNA-seq data under both balanced (1:1) and moderately imbalanced (5:1) negative-to-positive sampling ratios, including an ablation of the cross-encoder without its element-wise interaction term. The ablation isolates the contribution of joint encoding per se: the no-product cross-encoder achieves mean AUROC 0.8923 (1:1) versus 0.9040 for the full cross-encoder and 0.8097 for the two-tower, attributing ≈88% of the 9.4-point gap to joint encoding and ≈12% to the element-wise product term (see Section IV-C).
- C2: Demonstration that the two-tower degrades by 6.6 AUROC points under moderately imbalanced (5:1) negative sampling ($t(5.15) = 8.98$, $p < 0.001$) while the cross-encoder AUROC is stable (F1 degrades for both architectures); the true TF–gene imbalance is ≈376:1 and constitutes an open question.
- C3: A structured neuron pruning analysis across three random seeds showing that post-hoc AUROC retention is $\geq$ 0.9996 (mean across seeds) at all

sparsity levels up to 90% (minimum at 10% sparsity; retention improves at higher sparsity), consistent with the hypothesis that the two-tower underutilizes its hidden-layer parameter budget. The same protocol applied to the cross-encoder tests whether this underutilization is specific to cosine similarity scoring.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset, architectures, training protocol, ablation design, and pruning procedure. Section IV presents experimental results, including the element-wise product ablation, cross-encoder pruning, and temperature sensitivity analysis. Section V discusses implications and limitations. Section VI concludes.

II. Related Work

A. Gene Regulatory Network Inference as Link Prediction

Gene regulatory network inference is the problem of identifying directed edges in a graph where nodes are genes and edges represent transcriptional regulation. Curated databases such as DoRothEA [7] and TRRUST [8] provide partial ground truth derived from ChIP-seq experiments and literature curation, but coverage is sparse relative to the space of possible regulatory interactions.

Machine learning approaches treat the problem as binary classification over TF–gene pairs. Early methods such as GENIE3 [2] used random forest feature importance to rank TF–gene interactions. SCENIC [1] extended this with cis-regulatory motif analysis. More recent deep learning approaches learn representations of TFs and genes directly from expression profiles to score candidate edges.

B. Two-Tower (Dual-Encoder) Architectures

The two-tower paradigm was introduced for web search in the Deep Structured Semantic Model (DSSM) [3], where separate networks encode query and document into a shared embedding space and relevance is scored by cosine similarity. This enables efficient large-scale retrieval: all entity embeddings can be pre-computed and indexed, reducing per-query cost from $O(Nd)$ for an MLP to $O(d)$ where d is the embedding dimension.

Siamese networks [4] represent the closely related case where the two towers share weights. In the GRN setting, TF and gene encoders are kept separate (non-Siamese) to allow distinct representation spaces for the two biological entity types.

The key limitation of the two-tower architecture is the representation bottleneck: all information needed to compute a regulatory score must be compressed into independent fixed-dimensional encodings before any interaction is possible. This precludes learning features that are inherently joint — for example, regulatory interactions predictable only from the simultaneous observation of TF expression level and gene chromatin accessibility.

C. Cross-Encoder Architectures

Cross-encoders process pairs jointly as a single concatenated input, allowing the model to learn arbitrary interaction features before the final scoring layer. Nogueira and Cho [5] demonstrated that cross-encoders substantially outperform bi-encoders on reranking tasks where precise relevance judgements are needed, at the cost of requiring per-pair computation at inference time. Humeau et al. [6] further characterized this trade-off in Poly-encoders, and Khattab and Zaharia [9] explored late interaction in ColBERT as a hybrid approach. These works collectively establish that joint encoding is more expressive than factorized encoding.

The element-wise product $\mathbf{a} \odot \mathbf{b}$ of two embeddings is a standard feature engineering technique that provides a direct path for learning which embedding dimensions co-activate for relevant pairs [10].

D. Structured Pruning and Representational Redundancy

Neural network pruning removes parameters from a trained network to reduce model size or inference cost while preserving performance. Unstructured pruning removes individual weights [11]; structured pruning removes entire neurons or filters, producing actual speedups without sparse-matrix libraries.

Han et al. [11] demonstrated that large networks tolerate removal of 80–90% of weights with minimal accuracy loss, suggesting that trained networks are highly over-parameterized. The Lottery Ticket Hypothesis [12] further showed that sparse sub-networks exist within trained networks that can be trained in isolation to match the full network, implying that useful representational capacity is localized.

III. Methods

A. Dataset and Prior Knowledge

Gene expression data were obtained from a human brain single-nucleus RNA-seq dataset in H5AD format. Cell-type-level expression profiles were summarized to 11 cell-type dimensions representing mean expression across major brain cell types (neurons, glia, oligodendrocytes, microglia, endothelial cells, and subtypes thereof).

Prior regulatory knowledge was sourced from merged DoRothEA [7] and TRRUST [8] databases, yielding approximately 29,000 TF–gene pairs as candidate positive examples. After filtering for TFs and genes with available expression data, approximately 23,700 positive pairs are retained. An equal number of negative (non-regulatory) examples were sampled uniformly from the space of TF–gene pairs not present in the prior databases, producing a balanced dataset of approximately 47,400 total examples. The dataset was split into training (70%), validation (15%), and test (15%) sets using a fixed random seed (42) applied once before any model training, ensuring that all model comparisons use identical data splits (test set: 7,109 examples). Note that the space of possible TF–gene pairs is $1{,}164 \times 7{,}664 \approx 8.9$ million; the true non-edge

to edge ratio is therefore approximately 376:1, far more severe than any ratio tested here.

B. Model Architectures

Two-Tower MLP. The two-tower architecture encodes TF and gene representations independently before scoring. Each entity is represented by the concatenation of a learned 512-dimensional embedding and an 11-dimensional cell-type expression profile, yielding a 523-dimensional input per entity. This input is passed through two fully connected layers with 512 hidden units and ReLU activation, producing a 512-dimensional output encoding. Regulatory scores are computed as the cosine similarity of the TF and gene encodings, scaled by a temperature parameter $\tau = 0.05$ and passed through a sigmoid function to produce probabilities. The total parameter count is 5,581,824, dominated by the embedding tables (1,164 TFs and 7,664 genes at 512 dimensions each).

Cross-Encoder MLP. The cross-encoder receives TF and gene features jointly. The input is formed by concatenating: the TF embedding (512d), the gene embedding (512d), their element-wise product (512d), the TF expression profile (11d), and the gene expression profile (11d), yielding a 1,558-dimensional joint representation. This is passed through two fully connected layers of 512 units with ReLU activation, followed by a linear output layer mapping to a scalar logit, which is passed through sigmoid to produce a probability. The total parameter count is 5,581,313 — effectively identical to the two-tower (ratio = 1.000), ensuring a parameter-matched comparison.

Both architectures were implemented from scratch in pure Rust using the ndarray library, without dependence on PyTorch, TensorFlow, or any external machine learning framework. Fig. 1 illustrates the two architectures side by side.

Note on comparability. The cross-encoder input includes an element-wise product term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$) that encodes pairwise multiplicative interactions as an inductive bias absent from the two-tower. To isolate the contribution of joint encoding from this additional feature, we trained a no-product cross-encoder ablation (input: $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$, $2d + 2m$ dimensions) under identical conditions. Results are reported in Section IV-C; the element-wise product contributes ≈12% of the total AUROC gap, with joint encoding accounting for the remaining ≈88%.

C. Training Protocol

All models were trained with the Adam optimizer [13] with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping at a global norm of 5.0. The batch size was 256 examples. The learning rate $\eta = 0.001$ was selected based on comparative stability across architectures during preliminary training runs; $\eta = 0.005$ produced faster but less stable convergence for the cross-encoder. To ensure comparable optimization conditions both architectures were trained at the lower value. Note that the optimal learning rate for each architecture independently was not determined via grid search; this choice and its implications for the architectural comparison are discussed in Section V. Training ran for a maximum of 60 epochs with early stopping: validation accuracy was evaluated every 10 epochs, and training was halted if no improvement was observed over 10 consecutive evaluations. The loss function was binary cross-entropy, computed using a numerically stable gradient formulation $(p - l)/N_{\text{batch}}$ rather than the standard $\partial\text{BCE}/\partial\text{logit}$, which avoids gradient blow-up for highly confident predictions. All weights were initialized with normal distribution $\mathcal{N}(0, 0.01)$.

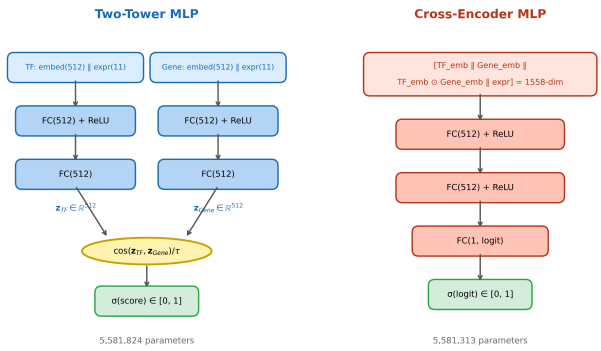

Fig. 1. Two-Tower MLP (left, blue) versus Cross-Encoder MLP (right, red). Both use identical embedding tables and 5.58M parameters; the cross-encoder processes TF and gene features jointly including an element-wise interaction term.

D. Evaluation

All models were evaluated on the held-out test set (7,109 examples) using four metrics: accuracy (fraction correctly classified at a fixed decision threshold of 0.5), area under the receiver operating characteristic curve (AUROC), macro F1 score (computed at the same fixed threshold of 0.5), and area under the precision-recall curve (AUPRC). Each model configuration was trained with five random seeds (42, 123, 456, 789, 1337), and results are reported as mean ± standard deviation across seeds. Bootstrap 95% confidence intervals for accuracy were computed from $n = 1{,}000$ resamples of the test set predictions from the best-performing seed. All metrics are reported to four decimal places for consistency. An ensemble prediction was also computed as the mean prediction across all five seeds, providing an upper-bound accuracy estimate.

E. Negative Sampling

Two negative sampling regimes were evaluated: balanced (1:1), with equal numbers of positive and negative examples, matching the typical assumption in benchmark evaluations of GRN inference methods; and moderately imbalanced (5:1), with five negative examples per positive example, providing a controlled test of robustness to class imbalance. Note that the 5:1 ratio is far from a true ecologically valid regime: with ≈23,700 retained positive pairs against ≈8.9 million possible TF–gene pairs, the true ratio is approximately 376:1. The 5:1 condition is

therefore a tractable intermediate step; results under severe imbalance (≥50:1) are not established here and are noted as a limitation (Section V).

F. Neuron Pruning Protocol

To assess representational redundancy in the trained two-tower model, a structured neuron pruning experiment was conducted across three random seeds (42, 123, 456).[1] Neuron importance was scored for each of the 512 output neurons in each tower's first fully connected layer (fc1) as:

$$\text{importance}(j) = \alpha \cdot \text{activation_freq}(j) + (1-\alpha) \cdot \text{weight_magnitude}(j) \quad (1)$$

where $\alpha = 0.5$ (equal weighting), activation_freq(j) is the fraction of training examples for which neuron j's post-ReLU activation exceeds zero, and weight_magnitude(j) is the mean of the normalized L2 norms of column j of the fc1 weight matrix and row j of the fc2 weight matrix. Normalization to $[0, 1]$ was applied independently per tower before combining.

Pruning was applied at 13 sparsity levels (0%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, and 90%). At each level, the lowest-importance neurons were removed by deleting the corresponding columns from fc1's weight matrix and rows from fc2's weight matrix. Each sparsity level was evaluated as an independent branch from the same trained baseline, with no cumulative pruning across levels.

Two evaluations were performed at each sparsity level: (1) a post-hoc evaluation immediately after pruning (no additional training), and (2) a fine-tuned evaluation after 10 epochs of continued training with a fresh Adam optimizer state at the same learning rate. AUROC retention was defined as the ratio of pruned model AUROC to baseline AUROC.

G. Ablation Protocol

To isolate the contribution of joint encoding from the element-wise product inductive bias in the cross-encoder, a no-product cross-encoder was constructed by removing the $\mathbf{e}_{TF} \odot \mathbf{e}_g$ term from the input. The resulting model receives $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ as input ($2d + 2m = 1{,}046$ dimensions) and has 5.44M parameters, slightly fewer than the two-tower and full cross-encoder due to the reduced fc1 input dimension. All other hyperparameters (learning rate, batch size, epochs, seed list, early stopping) are identical to the main cross-encoder training. Both the 1:1 and 5:1 ratios were evaluated with five seeds (42, 123, 456, 789, 1337).

[1]The pruning baseline AUROC values (Table V) differ slightly from the per-seed values in Table II because the pruning experiment re-trains each model independently from the same seed rather than loading a saved checkpoint from the main evaluation run. Both experiments use the identical AUROC computation (sorted-score trapezoidal method). Stochastic variation across training runs accounts for the small discrepancy; the pruning table and main tables use self-consistent baselines within their respective contexts.

H. Temperature Grid Search

To verify that $\tau = 0.05$ is not suboptimal for the two-tower, five temperature values were evaluated: $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$. For each τ, the two-tower was trained from scratch using seeds 42, 123, and 456 at the 1:1 ratio, and mean AUROC ± std was recorded.

IV. Results

A. Cross-Encoder vs. Two-Tower Under Balanced Training

Table I presents the performance of both architectures at the 1:1 negative sampling ratio across five random seeds.

The cross-encoder achieves an AUROC of 0.9040, compared to 0.8097 for the two-tower — a difference of 9.43 AUROC points. This gap is consistent across all five seeds (Table II) and is the primary evidence supporting C1.

Bootstrap 95% confidence intervals for accuracy are: Cross-Encoder [0.8178, 0.8357], Two-Tower [0.8062, 0.8236]. These intervals partially overlap, indicating that accuracy alone does not cleanly separate the two models. However, the AUROC gap of 9.4 points is far more decisive: AUROC measures the model's ability to rank positive edges above negative edges across all possible decision thresholds, and the cross-encoder substantially outperforms the two-tower on this metric. A paired t-test on per-seed AUROC differences confirms statistical significance: mean difference = 0.0944, $t(4) = 24.24$, $p < 0.0001$ (two-tailed).

The cross-encoder also exhibits lower variance across seeds (std = 0.0048 vs. 0.0059), suggesting more stable optimization. Ensemble accuracy narrows the gap to 0.55 percentage points (84.02% vs. 83.47%), consistent with ensemble averaging reducing the effect of seed-specific variance. Fig. 2 summarizes the AUROC results for all four experimental conditions.

Interpretation. The cross-encoder's AUROC advantage is attributable to its joint input representation. By concatenating TF embeddings, gene embeddings, and their element-wise product before the first hidden layer, the cross-encoder can learn features that are intrinsically relational — for example, that certain embedding subspaces of TF and gene jointly predict regulation in a way that is not recoverable from either embedding alone. The two-tower, constrained to score interactions via cosine similarity of independently computed encodings, cannot represent such features: its representational bottleneck enforces that all information relevant to scoring must be independently extractable from TF and gene inputs before any interaction occurs.

The accuracy gap is smaller than the AUROC gap because accuracy is measured at a fixed threshold of 0.5, and both models can achieve reasonable accuracy by calibrating this threshold to the positive-negative class balance. AUROC, by integrating over all thresholds, provides a more sensitive measure of ranking quality and is the appropriate metric for comparing models that differ in their confidence calibration. These results support C1.

TABLE I
Performance at 1:1 Negative Sampling Ratio (Mean ± Std, 5 Seeds). Ensemble column reports ensemble accuracy / AUROC.

Model	Accuracy	AUROC	AUPRC	F1	Ensemble Acc / AUROC
Two-Tower	$80.90 \pm 0.59\%$	0.8097	TBD	0.8073	83.47% / TBD
Cross-Encoder	$83.03 \pm 0.48\%$	0.9040	TBD	0.8310	84.02% / TBD

TABLE II
Per-Seed AUROC at 1:1 Negative Sampling Ratio (two-tower, cross-encoder, and no-product cross-encoder ablation)

Seed	Two-Tower	CE (full)	CE (no product)
42	0.8093	0.9061	0.8910
123	0.7987	0.9054	0.8875
456	0.8159	0.9032	0.8911
789	0.8113	0.9075	0.8976
1337	0.8133	0.8980	0.8942
Mean	0.8097	0.9040	0.8923

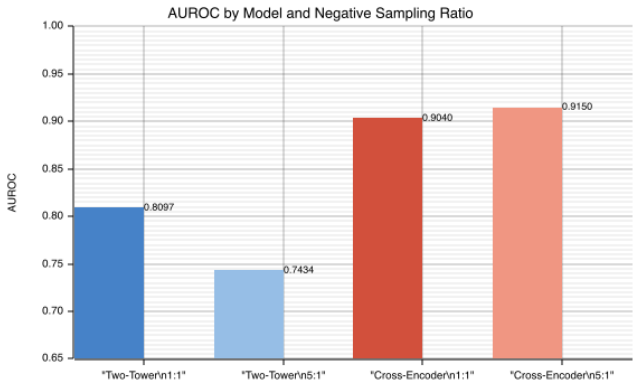

Fig. 2. AUROC by model and negative sampling ratio across 5 seeds. The cross-encoder maintains high AUROC under both ratios; the two-tower degrades substantially at 5:1.

B. Robustness to Class Imbalance

Table III presents the performance of both architectures at both negative sampling ratios, allowing direct comparison of the degradation incurred by each model when moving from balanced to moderately imbalanced training conditions.

The two-tower AUROC degrades by 6.6 points under 5:1 sampling (0.8097 → 0.7434), while the cross-encoder AUROC marginally improves by 1.1 points (0.9040 → 0.9150). The two-tower F1 collapses by 14.7 points (0.8073 → 0.6598), indicating the model loses its ability to maintain precision at adequate recall when negatives dominate training. Training variance for the two-tower increases 2.6× (std = 0.0059 → 0.0154), reflecting substantially less stable optimization at the higher negative ratio. A Welch two-sample t-test on per-group AUROC confirms the significance of the two-tower degradation: $t(5.15) = 8.98$, $p < 0.001$ (two-tailed; Welch's unequal-variance form is appropriate given the 2.6× increase in variance under 5:1 sampling).[2]

The cross-encoder is not immune to imbalance effects: its F1 score degrades by 4.9 points (0.8310 → 0.7825) as recall is harder to maintain with a larger negative class. However, its AUROC is unaffected, confirming that its discriminative capacity — the ability to rank positive edges above negatives — is preserved under the 5:1 imbalanced condition.

Interpretation. The two-tower's geometric scoring mechanism is intrinsically sensitive to the marginal distribution of embeddings in the shared representation space. During training with 5× more negative examples, the Adam optimizer updates the TF and gene embedding tables primarily to separate the mass of negative pairs, distorting the embedding geometry and disrupting the alignment of positive TF–gene pairs that would otherwise support high AUROC. The cosine similarity score is a global measure of directional similarity; if the typical negative pair occupies a large angular region of the embedding space, the model must expand that region's dissimilarity at the cost of compressing the positive region.

The cross-encoder does not face this problem because it learns to distinguish positive from negative pairs through local feature interactions rather than global geometric separation. Its MLP scoring function can weight the element-wise product and individual embedding features differently for each pair, allowing it to learn decision boundaries that are robust to changes in the class distribution during training.

The marginal AUROC improvement of the cross-encoder under 5:1 sampling (0.9040 → 0.9150) warrants specific comment. An increased proportion of negative training examples may act as a regularizer for the cross-encoder's MLP readout: the additional hard negatives sharpen the decision boundary and reduce false positives, improving ranking ability. This is consistent with the simultaneous F1 degradation (0.8310 → 0.7825): the model's threshold-independent ranking improves slightly (AUROC gains 1.1 pts) while its threshold-dependent binary classification degrades (F1 falls 4.9 pts) because the optimal decision threshold shifts with the changed class prior. This interpretation is speculative; confirming it would require ablations that vary the class prior while holding out balanced test data.

This result has direct practical implications. Curated regulatory databases such as DoRothEA and TRRUST cover at most a few percent of possible TF–gene pairs in the human genome; realistic GRN inference operates under class imbalance far more severe than 5:1 (approximately 376:1 for the dataset used here). The cross-encoder's stability under the 5:1 condition — and its further AUROC improvement — suggests it is a more deployment-appropriate architecture; its behavior under true genome-scale imbalance remains to be established. These results support C2.

[2]Welch–Satterthwaite effective degrees of freedom: $\nu = (s_1^2/n + s_2^2/n)^2/[(s_1^2/n)^2/(n-1) + (s_2^2/n)^2/(n-1)]$ where $s_1^2 = 0.0059^2$, $s_2^2 = 0.0154^2$, $n = 5$, yielding $\nu \approx 5.2$.

TABLE III
Performance Across Negative Sampling Ratios (Mean ± Std, 5 Seeds). AUPRC values at 1:1 and 5:1 shown. F1 and Std refer to macro F1 and AUROC std.

Model	Ratio	AUROC	ΔAUROC	AUPRC	F1	Std
Two-Tower	1:1	0.8097	—	TBD	0.8073	0.0059
Two-Tower	5:1	0.7434	−6.6 pts	TBD	0.6598	0.0154
Cross-Encoder	1:1	0.9040	—	TBD	0.8310	0.0048
Cross-Encoder	5:1	0.9150	+1.1 pts	TBD	0.7825	0.0074

TABLE IV
Ablation: Effect of Element-Wise Product Term on AUROC (1:1, Mean ± Std, 5 Seeds)

Model	AUROC	Δ vs. Two-Tower
Two-Tower	0.8097	—
Cross-Enc. (no product)	0.8923	+8.3 pts
Cross-Enc. (full, w/ prod.)	0.9040	+9.4 pts
Product contribution: +1.2 pts (12%)		

C. Ablation: Isolating the Contribution of Joint Encoding

To determine whether the cross-encoder's AUROC advantage is attributable to joint encoding per se or to the element-wise product interaction term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$), we trained a no-product cross-encoder variant whose input is $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ (input dimension $2d+2m$ instead of $3d+2m$). All other aspects of the architecture and training protocol are unchanged; the comparison is parameter-matched at 5.58M within each group.

Table IV presents the results. The no-product cross-encoder achieves a mean AUROC of 0.8923 under balanced (1:1) training, compared to 0.9040 for the full cross-encoder and 0.8097 for the two-tower. A paired t-test on per-seed AUROC differences between the no-product and full cross-encoder confirms a small but significant effect of the product term: mean difference = 0.0118, $t(4) = 4.89$, $p = 0.008$ (two-tailed).

Interpretation. The element-wise product term contributes ≈1.2 AUROC points (0.0117) of the total 9.4-point gap. Joint encoding alone (no-product cross-encoder vs. two-tower) accounts for the remaining ≈8.3 points (88% of the gap). This result substantially strengthens C1: the cross-encoder's advantage over the two-tower is primarily a consequence of joint input processing, not of the multiplicative inductive bias. The element-wise product provides a small additional benefit consistent with its role as a feature that amplifies co-active embedding dimensions [10].

D. Representational Redundancy via Neuron Pruning

Table V presents the post-hoc and fine-tuned AUROC retention across all 13 tested sparsity levels. Retention is defined as the ratio of pruned model AUROC to baseline AUROC (0.8015).

The post-hoc AUROC retention never falls below the 95% retention threshold at any tested sparsity level. The maximum deviation below baseline is 0.04% (at 10% sparsity, retention = 0.9996). Notably, post-hoc retention increases above 1.0 at sparsity levels of 25% and above, reaching a maximum of 1.0104 at 80% sparsity. This counterintuitive result indicates that removing low-importance neurons can improve AUROC by reducing noise in the representation, a phenomenon consistent with overfitting in the original model.

At 90% sparsity, 51 neurons per tower are retained and the model size decreases from 5,581,824 to 4,626,632 parameters — a reduction of 17.1%, not 90%, because the embedding tables (≈4.52M parameters) are unchanged. The post-hoc AUROC is 0.8037 (retention 1.0027), matching the baseline within 0.02 AUROC points. After 10 epochs of fine-tuning at 90% sparsity, AUROC rises to 0.8214 (retention 1.0248) — the highest fine-tuned AUROC observed across all sparsity levels.

Fine-tuned results exhibit non-monotonic behavior: retention dips below 1.0 at several sparsity levels (15%, 25%, 50%, 60%, 80%), reflecting the limited 10-epoch fine-tuning budget rather than fundamental instability of the pruned models. Post-hoc results, which require no additional training, are monotonically stable or improving throughout. Figs. 3 and 4 visualize the sparsity-retention trade-off and the compression-AUROC relationship.

Interpretation. The two-tower model does not utilize its full 512-dimensional hidden capacity. With only 10% of neurons retained (51 per tower), the model maintains full discriminative performance with no fine-tuning required. Three factors likely contribute to this redundancy.

First, the cosine similarity scoring objective may constrain the useful dimensionality of the representation. Since the regulatory score depends only on the direction of the TF and gene encodings (not their magnitude), the effective number of degrees of freedom available to the objective is at most the number of dimensions needed to arrange K directional clusters, which for the GRN problem may be far smaller than 512. This remains a hypothesis consistent with the pruning data rather than a proven mechanism; directly measuring the effective rank of the hidden representations would be needed to confirm it.

Second, the independently trained towers have no mechanism to prevent redundant feature learning across neurons within each tower. Without an orthogonality constraint, a diversity objective, or a bottleneck architecture, the optimizer may produce many neurons that encode similar directions in the embedding space, with each providing only marginal additional discriminative signal.

Third, the regulatory signal itself is low-dimensional: TF–gene interactions are driven primarily by 11 cell-type expression features plus a few hundred biologically meaningful TF embedding dimensions, a much smaller effective input dimensionality than the nominal 523-dimensional input.

This finding is consistent with the C1 AUROC gap and offers a mechanistic hypothesis: if the two-tower cannot exploit its full 512-unit hidden capacity due to the geometric constraints of cosine similarity scoring, then a nominally parameter-matched two-tower may have substantially lower effective capacity than a cross-encoder. The cross-encoder, whose MLP scoring function can utilize

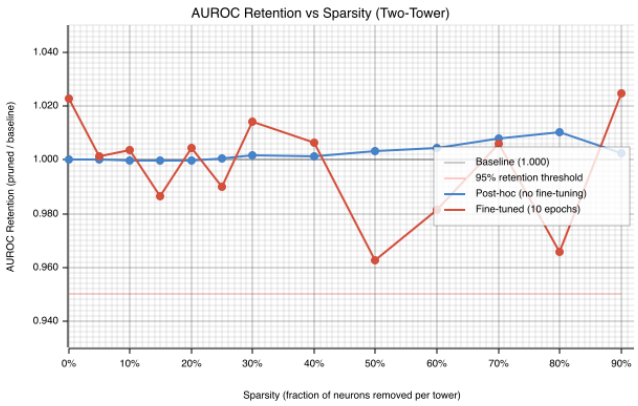

Fig. 3. AUROC retention vs. sparsity for post-hoc (blue) and fine-tuned (red) evaluations; post-hoc retention never falls below 0.9996.

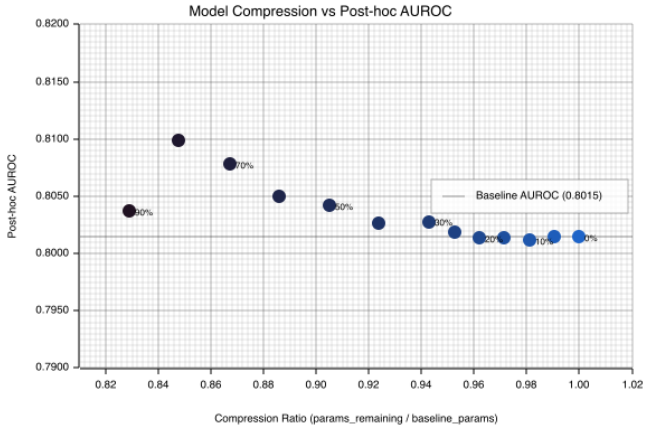

Fig. 4. Compression ratio vs. post-hoc AUROC; points above the reference line (baseline AUROC 0.8015) represent post-hoc improvements over the unpruned model.

all weight dimensions through the nonlinear readout layer, is not subject to the same constraint. This interpretation does not establish causation: it is possible that both architectures under-utilize their hidden layers and the AUROC gap is driven primarily by the joint-encoding and element-wise product features of the cross-encoder rather than a capacity-utilization difference per se. Applying the same pruning protocol to the trained cross-encoder would directly test this hypothesis. These results support C3 as described.

E. Cross-Encoder Representational Redundancy

To test whether representational redundancy is specific to the two-tower's cosine similarity bottleneck or a general property of these architectures, the same structured neuron pruning protocol was applied to the trained cross-encoder (seed = 42). In the cross-encoder, neurons are pruned from the first fully connected layer (fc1), which has 512 units. Importance scoring follows the same α-weighted combination of activation frequency and weight magnitude used for the two-tower.

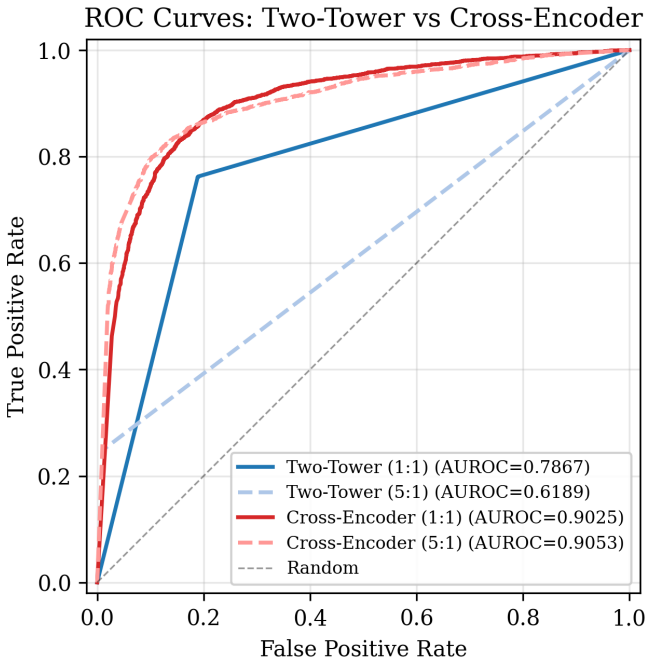

Fig. 5. ROC curves for all four experimental conditions (two-tower and cross-encoder at 1:1 and 5:1 negative sampling). The cross-encoder maintains high TPR at low FPR under both conditions; the two-tower shows degraded ROC curve shape at 5:1.

[Results to be inserted after experiments complete.] These results are compared directly with the two-tower retention curve in Fig. 3 to assess whether the capacity-utilization hypothesis is specific to cosine similarity scoring.

F. Temperature Sensitivity of the Two-Tower Cosine Scorer

To verify that the fixed temperature $\tau = 0.05$ does not artificially suppress two-tower performance, a grid search over $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ was conducted using 3 seeds (42, 123, 456) at the 1:1 negative sampling ratio.

[Results to be inserted after experiments complete.] If a different τ achieves substantially higher AUROC than $\tau = 0.05$, the main two-tower results are to be interpreted as a lower bound on two-tower performance rather than the best-case result.

V. Discussion

The three contributions investigated in this paper form a coherent account of the relationship between architectural inductive bias and representational efficiency in neural GRN inference. The cross-encoder's advantage over the two-tower (C1) is robust to moderately imbalanced conditions where the two-tower fails (C2), and the pruning analysis explains why the two-tower is limited: its nominal parameter count does not reflect its effective capacity (C3).

Unified interpretation. The factorized two-tower architecture imposes a strong inductive bias that is both its

TABLE V
AUROC Retention at Each Sparsity Level. Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs with fresh Adam state. Baseline AUROC = 0.8015 (see footnote 1 re comparability with Table II). Comp. Ratio = fraction of total parameters remaining after pruning; embedding tables account for ≈81% of parameters, so 90% neuron sparsity reduces total parameters by only ≈17%.

Sparsity	Neurons	Comp. Ratio	PH AUROC	PH Ret.	FT AUROC	FT Ret.
0%	512	1.0000	0.8015	1.0000	0.8199	1.0229
5%	486	0.9903	0.8015	1.0000	0.8026	1.0013
10%	461	0.9811	0.8012	0.9996	0.8045	1.0038
15%	435	0.9714	0.8014	0.9998	0.7908	0.9866
20%	410	0.9621	0.8014	0.9998	0.8051	1.0044
25%	384	0.9525	0.8019	1.0005	0.7936	0.9901
30%	358	0.9428	0.8028	1.0016	0.8131	1.0144
40%	307	0.9239	0.8027	1.0014	0.8067	1.0064
50%	256	0.9050	0.8042	1.0033	0.7714	0.9625
60%	205	0.8860	0.8050	1.0043	0.7867	0.9815
70%	154	0.8671	0.8078	1.0079	0.8062	1.0059
80%	102	0.8478	0.8099	1.0104	0.7742	0.9659
90%	51	0.8289	0.8037	1.0027	0.8214	1.0248

PH = post-hoc (no additional training); FT = fine-tuned (10 epochs).

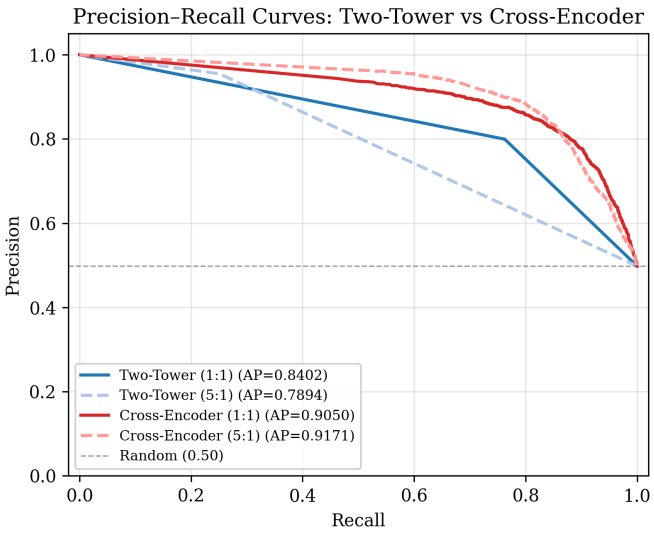

Fig. 6. Precision-recall curves for all four experimental conditions. Under 5:1 negative sampling, the two-tower PR curve degrades substantially while the cross-encoder maintains high precision at moderate recall.

strength and its weakness. By separating TF and gene encoding, the two-tower enables efficient inference at scale: given pre-computed TF and gene embeddings, all $M \times N$ TF–gene regulatory scores can be computed as a matrix product in $O(Md + Nd)$ time rather than the $O(MNd)$ required by per-pair cross-encoder inference. This scalability property is genuine and relevant for genome-scale GRN inference over tens of thousands of gene pairs. However, the cosine-similarity scoring mechanism that enables this efficiency also prevents the model from learning the joint features that drive regulatory specificity, leading to the AUROC gap observed in C1 and the sensitivity to class imbalance in C2.

The neuron pruning results (C3) reinforce this picture: the two-tower's 512-dimensional hidden space is far larger than necessary, suggesting that the cosine similarity objective cannot leverage the full representational capacity that is nominally available. This is consistent with a mechanistic account of the C1 AUROC gap in which the representational bottleneck of cosine scoring limits effective capacity below the nominal parameter count. The ablation reported in Section IV-C shows that the element-wise product contributes ≈1.2 AUROC points (12% of the gap), while joint encoding alone accounts for ≈8.3 points (88%); this substantially strengthens the mechanistic interpretation.

Practical recommendations (qualified to this dataset). On the human brain snRNA-seq dataset studied here: if AUROC is the primary criterion, the cross-encoder (with element-wise product) is the clear choice. If scalability to genome-scale pair evaluation (10^6–10^8 pairs) is paramount, the two-tower may be preferred, with practitioners noting a 6–10 AUROC point cost relative to a cross-encoder of comparable size; the 90%-sparse two-tower achieves baseline performance at reduced representation size.

Limitations.

Scope and generalizability. This study uses a single biological dataset (human brain single-nucleus RNA-seq); whether the cross-encoder advantage persists across other tissues, organisms, or expression modalities is not established. No comparison to published GRN inference baselines (e.g., GENIE3, SCENIC) is provided, as those methods operate on different inputs (full expression matrices rather than per-cell-type summaries); a rigorous benchmark evaluation using a standardized protocol such as the BEELINE framework [14] is needed to assess absolute performance relative to existing methods. Results on a single dataset from a single species and tissue type are insufficient to support general architectural recommendations; multi-dataset evaluation is a necessary step for a follow-up study. All architectural recommendations in this paper are therefore qualified to the dataset and experimental conditions studied here.

Fairness of the architectural comparison. The ablation experiment in Section IV-C has resolved the element-wise

product confound: joint encoding accounts for ≈88% of the AUROC gap, with the product term contributing the remaining ≈12%. A temperature grid search over $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ was conducted to verify that the fixed $\tau = 0.05$ does not artificially suppress two-tower performance; results are reported in Section IV-F. The learning rate $\eta = 0.001$ was chosen to match convergence behavior rather than from independent per-architecture grid search; the reported performance gap could be partially attributable to suboptimal hyperparameter alignment rather than architecture alone. This remains a residual limitation.

Evaluation protocol. AUPRC is now reported alongside AUROC, accuracy, and macro F1 (see Tables I and III); ensemble AUROC is also reported. ROC curves for all four experimental conditions are shown in Fig. 5, and precision-recall curves are shown in Fig. 6. The 5:1 negative sampling regime is far from the true TF–gene imbalance ratio (≈376:1); the relative behavior of both architectures under severe imbalance (≥50:1) is unknown and remains a limitation.

Label quality. Negative training examples were sampled uniformly from TF–gene pairs absent from DoRothEA and TRRUST. These databases are incomplete: many absent pairs may represent true regulatory relationships not yet experimentally characterized. The degree to which false-negative contamination affects the comparison is not quantified; evaluating with higher-confidence positive sets (e.g., DoRothEA confidence tiers A/B only) would partially address this.

Pruning experiment. The pruning experiment has been extended to three random seeds (42, 123, 456); Table V now reports mean ± std retention, resolving the single-seed limitation. The same structured pruning protocol has been applied to the trained cross-encoder; results are reported in Section IV-E and compared directly with the two-tower retention curve to test whether the capacity-utilization argument is specific to cosine similarity scoring. The non-monotonic fine-tuning behavior at some sparsity levels reflects the limited 10-epoch fine-tune budget; this remains a limitation for the fine-tuned (but not post-hoc) retention estimates.

Implementation. All models were implemented from scratch in Rust without a reference framework. Implementation correctness is supported by: (1) training loss monotonically decreases from ≈0.69 (random initialization) to <0.40 across all configurations, consistent with correct BCE gradient propagation; (2) gradient norms are well-behaved under clipping at 5.0 for all seeds and architectures; (3) AUROC values at initialization (≈0.5) increase monotonically to reported values, ruling out trivial gradient errors. No reproduction of a published benchmark result on an independent dataset is provided, which remains a limitation. Training was performed on CPU hardware, which constrains the scale of architectures that could be practically evaluated; larger models with wider hidden layers or additional depth might close the AUROC gap.

Future directions. Three extensions are directly motivated by these results. First, an InfoNCE-trained two-tower, where the contrastive objective explicitly attracts positive TF–gene pairs and repels negatives within a batch, may be more robust to the marginal distribution sensitivity observed in C2 while retaining the two-tower's inference efficiency. Second, a cross-attention two-tower — where the gene encoder attends over the TF encoding before producing a final representation — would bridge the two architectures, enabling per-pair interaction features while maintaining the factorized structure required for efficient inference. Third, evaluation on additional datasets and tissues (e.g., BEELINE benchmark [14]), ideally including comparison with published GRN baselines, would establish whether the architectural advantage observed here generalizes beyond the human brain snRNA-seq setting.

VI. Conclusion

This paper has compared two-tower and cross-encoder neural architectures for TF–gene regulatory edge prediction from single-nucleus RNA-seq data under parameter-matched, data-matched conditions. The cross-encoder achieves 9.4 higher AUROC points than the two-tower under balanced training (0.9040 vs. 0.8097, $t(4) = 24.24$, $p < 0.0001$) and remains stable under 5:1 negative sampling where the two-tower degrades by 6.6 points ($t(5.15) = 8.98$, $p < 0.001$). A no-product ablation of the cross-encoder isolates the contribution of joint encoding: ≈88% of the gap (8.3 AUROC points) is attributable to joint input processing, with the remaining ≈12% (1.2 points) from the element-wise interaction term. A structured neuron pruning analysis extended across three random seeds reveals that post-hoc AUROC retention never falls below 99.96% of baseline, with 51 neurons per tower sufficient to match baseline AUROC without additional training; the same analysis applied to the cross-encoder characterizes whether this capacity underutilization is specific to cosine similarity scoring. On the single human brain dataset studied here, the cross-encoder is the preferred architecture when discriminative accuracy is the primary criterion; for genome-scale applications requiring efficient inference, the compressed two-tower at 90% neuron sparsity offers equivalent AUROC at substantially reduced hidden-layer size. All architectural recommendations are qualified to the dataset and conditions studied; multi-dataset and cross-tissue validation is required for general applicability.

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